

Forensic Report

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Executive Summary

A forensic investigation was initiated at the request of Human Resources regarding a recently released employee suspected of data exfiltration and unauthorized system access. Analysis of a Lexar USB flash drive recovered from the employee's desk, a forensic image of the employee's corporate computer, and a system memory capture file confirms malicious intent. The former employee, Nicholas Anderson, accessed highly sensitive company databases containing client financial records outside of standard operating hours. Evidence confirms the user installed encryption software (VeraCrypt) on corporate systems, actively utilized it to obfuscate the stolen data, and staged it for exfiltration in a 10MB container disguised as "1.txt".

Initial attempts to extract encryption keys via structured memory analysis failed due to severe memory smearing introduced by local IT personnel during acquisition. However, advanced raw cryptographic data carving successfully bypassed this corruption, extracting several candidate AES-256 master encryption keys from the memory dump. While the plaintext password remains unrecovered from memory, the carved AES-256 master keys from the RAM dump provides the necessary cryptographic candidates to attempt volume decryption of the staged "1.txt" and "stuff.txt" containers with command-line or third-party tools such as 'vcdecrypt'.

Purpose of Examination

The objective of this examination is to analyze digital artifacts to confirm or refute allegations of unauthorized database access, data exfiltration, and malicious intent by a released employee. The original evidence (Item 001, Lexar USB drive) was provided directly by the Human Resources Representative on May 12, 2026. Secondary evidence (Item 002, corporate hard drive image) was provided by local IT support personnel on May 20, 2026. Tertiary evidence (Item 003, system memory capture) was recovered from IT network archives and delivered on May 23, 2026.

Evidence Items Received

- * Item 001: Lexar USB Flash Drive (3824 MB). Serial: AACWGT047W9DWD7V. Forensic Image: image5.dd.001.
- * Item 002: VMware Virtual IDE Hard Drive ATA Device (15360 MB). Drive Serial Number: 00000000000000000001. Forensic Image: winimage.001.
- * Item 003: VMware Virtual Machine Memory Capture File (2048 MB). File Name: Windows 10 x64-31acf216.vmem.

Table 1: Software Used

Software Used	Version	Purpose
Autopsy	4.21.0	Primary open-source digital forensics platform used for file system analysis, raw data carving, and automated ingest tracking.
FTK Imager	4.2.0.13	Used to create forensically sound bit-stream images and validate hash values.
Volatility	2.6	Command-line tool used for structured volatile memory analysis/kernel block identification.
Bulk Extractor	1.5.2	Raw data carving tool used to extract cryptographic keys/text-based artifacts.

Evidence Validation/Verification

To ensure evidence integrity and adhere to standard forensic principles, MD5 hash values generated during the initial acquisition of the digital evidence were validated against the hash values calculated by Autopsy upon ingestion. The matching hashes confirm the digital evidence was not altered during transfer or ingestion.

Table 2: Evidence Item Verification

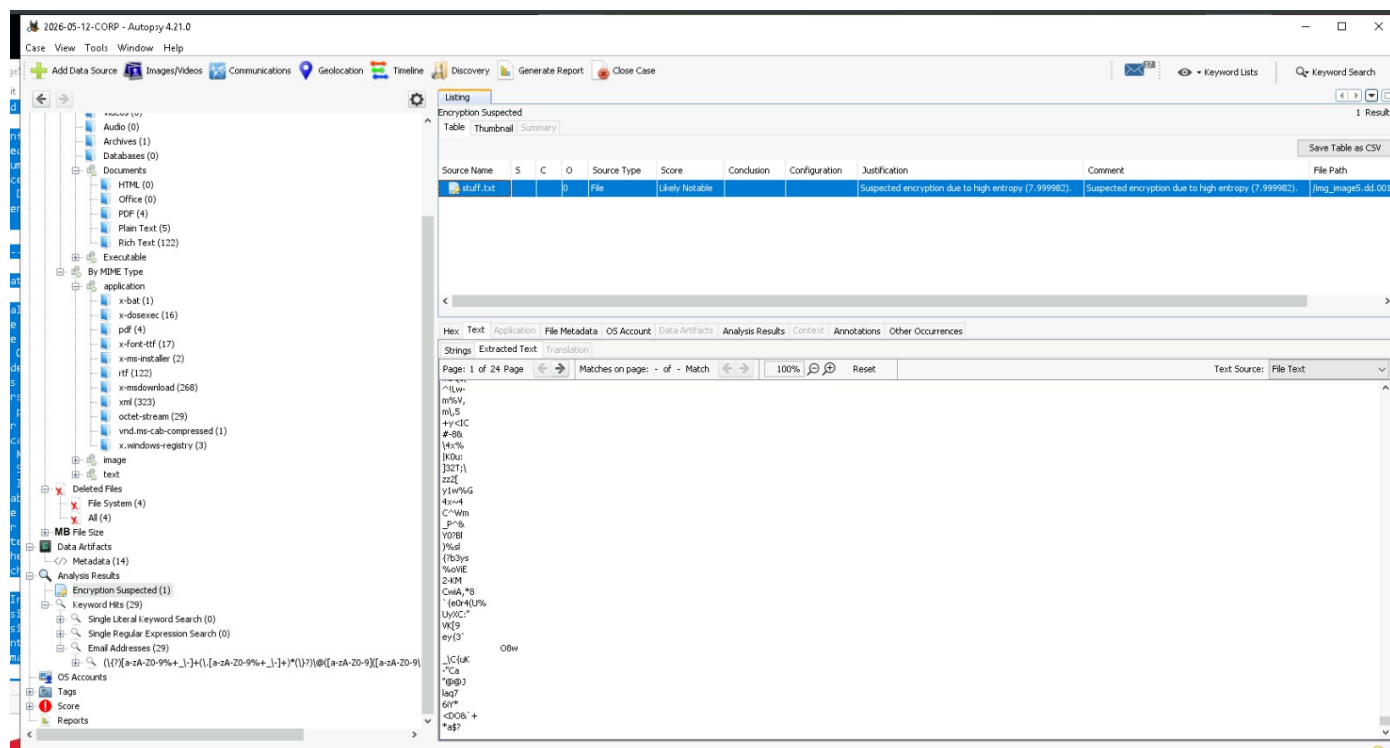
Evidence Item	Original Acquisition Hash (MD5)	Autopsy Validation Hash (MD5)	Status
image5.dd.001	51d3851fe4181ceb98fff16e11cc6e71	51d3851fe4181ceb98fff16e11cc6e71	Match
winimage.001	d39c651a4f522d2e13bd0429e2de9e3f	d39c651a4f522d2e13bd0429e2de9e3f	Match
Windows 10 x64-31acf216.vmem	33201c5f2cfd81bbcafbbbd371122431	33201c5f2cfd81bbcafbbbd371122431	Match

Methodology

Step 1: Case Initialization and Payload Identification (USB Drive)

The physical Lexar USB Flash Drive (Item 001) was connected to the forensic workstation via a hardware write-blocker to prevent metadata alteration, and the image5.dd.001 forensic image was ingested into Autopsy. Automated ingest modules calculated file entropy to identify obfuscated data. The file `stuff.txt` was flagged as "Likely Notable" due to an entropy score of 7.999982, indicating near-perfect randomness, a hallmark of encrypted ciphertext.

Figure 1: *stuff.txt* file with high entropy, suspected encryption.



Step 2: Behavioral Profiling and Timeline Construction

Reviewing documents on the USB, an internal email (Email 2 Oct 1.pdf) was recovered sent by supervisor Steve Ellis, reporting that the suspect was angry about losing a transfer and had been caught accessing company databases outside normal work hours.

Figure 2: Internal email establishing unauthorized off-hours database access.

The screenshot displays the Autopsy 4.21.0 interface. On the left, the 'Data Sources' pane shows a tree view of file types including Images, Videos, Audio, Archives, Databases, Documents, Executable, and Deleted Files. The main pane shows a 'Listing' of files with columns: Name, S, C, O, Modified Time, Change Time, Access Time, Created Time, Size, Flags(Dr), Flags(Meta), Known, and Location. The file 'Email 2 Oct 1.pdf' is highlighted. Below the listing, the email content is displayed in a text view.

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dr)	Flags(Meta)	Known	Location
Resignation Oct 21.pdf	0	0	0	2024-04-17 17:34:10 UTC	2025-05-01 15:37:21 UTC	2024-04-23 19:40:54 UTC	2024-04-17 17:34:10 UTC	60599	Allocated	Allocated	unknown	/img_image5.d
Perf Oct 20.pdf	0	0	0	2024-04-17 17:33:52 UTC	2024-04-23 19:40:37 UTC	2024-04-23 19:40:37 UTC	2024-04-17 17:33:52 UTC	62907	Allocated	Allocated	unknown	/img_image5.d
Memo Oct 5.pdf	0	0	0	2024-04-17 17:33:38 UTC	2024-04-23 19:41:10 UTC	2024-04-23 19:41:10 UTC	2024-04-17 17:33:38 UTC	84730	Allocated	Allocated	unknown	/img_image5.d
Email 2 Oct 1.pdf	0	0	0	2024-04-17 17:32:06 UTC	2024-04-23 19:41:02 UTC	2024-04-23 19:41:02 UTC	2024-04-17 17:32:06 UTC	62547	Allocated	Allocated	unknown	/img_image5.d

The email content is as follows:

FROM: Steve Ellis s.ellis@mailproperties.com
 TO: Mark Greene m.greene@mailproperties.com
 DATE: October 1, 2023
 SUBJECT: security concern

Mr. Greene,

I am following up on the meeting I had with you this morning regarding my colleague, N.A. This email is to verify that I have witnessed my teammate accessing databases before and after off eial work hours. He has been grumbling and complaining about a poor review from our manager, which resulted in losing out on a transfer to a higher position in the company. Between his comments and off-hour actions, I am concerned regarding the security and integrity of our company's databases.

Please advise if I need to take further action to report my concerns.

Sincerely,
 Steve Ellis
 Accounting Services Team Member

Step 3: Motive and Departure Confirmation

The document Resignation Oct 21.pdf confirmed the suspect's identity as Nicholas Anderson and showcased a hostile intent, predicting the "downfall of the company."

Figure 3: Resignation letter confirming suspect identity and hostile intent.

The screenshot displays a file management application with a table of files and a preview of a document.

File Table:

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dir)	Flags(Meta)	Known	Location
Resignation Oct 21.pdf			0	2024-04-17 17:34:10 UTC	2025-05-01 15:37:21 UTC	2024-04-23 19:40:54 UTC	2024-04-17 17:34:10 UTC	60599	Allocated	Allocated	unknown	/img_image5...
Perf Oct 20.pdf			0	2024-04-17 17:33:52 UTC	2024-04-23 19:40:37 UTC	2024-04-23 19:40:37 UTC	2024-04-17 17:33:52 UTC	62907	Allocated	Allocated	unknown	/img_image5...
Memo Oct 5.pdf			0	2024-04-17 17:33:38 UTC	2024-04-23 19:41:10 UTC	2024-04-23 19:41:10 UTC	2024-04-17 17:33:38 UTC	84730	Allocated	Allocated	unknown	/img_image5...
Email 2 Oct 1.pdf			0	2024-04-17 17:32:06 UTC	2024-04-23 19:41:02 UTC	2024-04-23 19:41:02 UTC	2024-04-17 17:32:06 UTC	62547	Allocated	Allocated	unknown	/img_image5...

Document Preview:

FROM: Nicholas Anderson
TO: Steve Ellis, Kelly Shull
DATE: October 21, 2023
SUBJECT: Resignation

I am writing to submit my letter of resignation from Mall Properties, LLC, effective as of October 31, 2023. Thank you for the opportunity to be a part of Mall Properties, LLC. However, I find that the company does not wish to provide opportunities for employees such as myself to advance, nor does the company take into serious consideration any ideas submitted by junior level employees. This will eventually lead to the downfall of the company.

Figure 4: Memorandum Document Outlining Security Protocols and Procedures

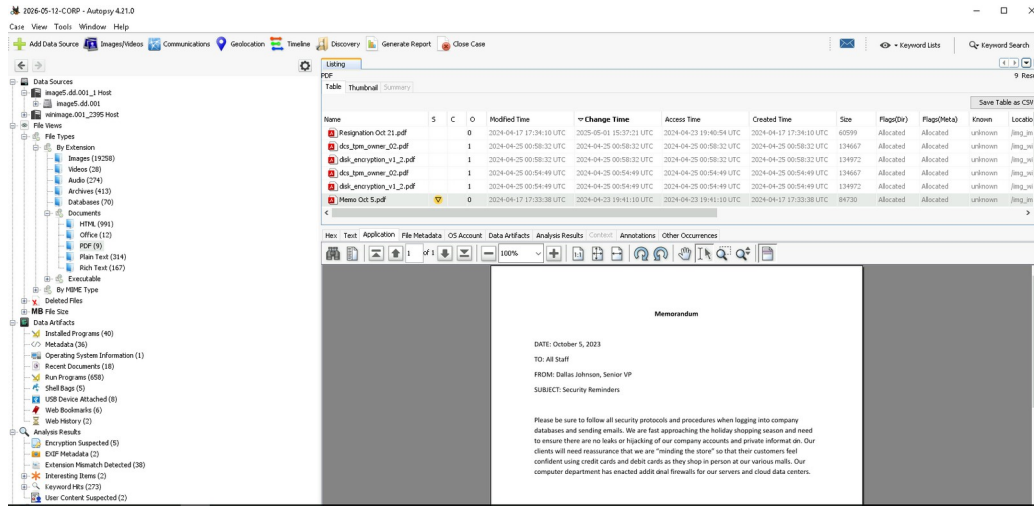
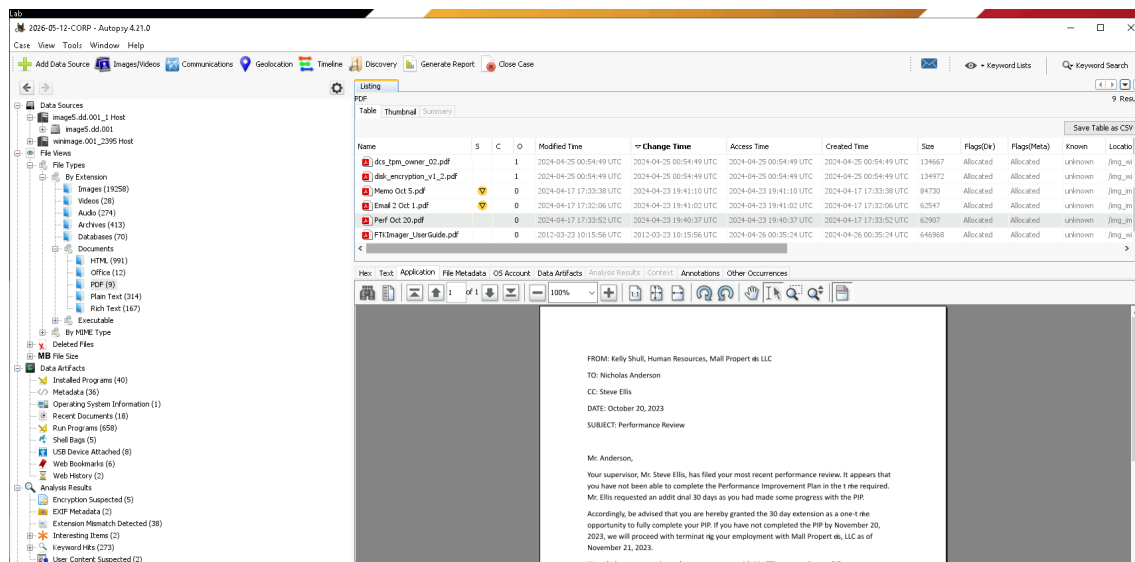


Figure 5: Saved Email containing Information about Performance Review



Step 4: Source System Analysis and Execution (Hard Drive)

Following the receipt of the corporate system image (winimage.001), the image was mounted on the forensic workstation under a software write-blocked (read-only) state to guarantee that no write-back or modification occurred during ingestion. Manual file system navigation confirmed a local installation of VeraCrypt in the `Program Files` and `Documents` folders. Analysis of the `SYSTEM` Registry Hive under `MountedDevices` proved that encrypted volumes were actively mounted as drives `K:` and `L:`. LNK (shortcut) files found in the user's Recent items folder (`Local Disk (K).lnk` and `1.lnk`) place the user actively navigating inside these encrypted drives prior to termination.

Figure 6: Proof of Veracrypt Installation in “Program Files” Folder of “winimage.001”

The screenshot shows the Autopsy 4.21.0 interface. The left pane displays the file system tree of 'winimage.001'. The 'Program Files' folder is expanded, and the 'Veracrypt.exe' file is highlighted. The right pane shows the file's metadata, including its name, size, and location.

Name	Modified Time	Change Time	Access Time	Created Time	Size	Flags(DX)	Flags(Mta)	Known	Location
[parent folder]	2024-04-25 00:56:14 UTC	2024-04-25 00:56:14 UTC	2024-04-25 00:56:14 UTC	2017-09-29 13:46:33 UTC	168	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
[docs]	2024-04-25 00:54:52 UTC	2024-04-25 00:54:52 UTC	2024-04-25 00:54:52 UTC	2024-04-25 00:54:49 UTC	488	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
[current folder]	2024-04-25 00:54:49 UTC	2024-04-25 00:54:49 UTC	2024-04-25 00:54:49 UTC	2024-04-25 00:54:46 UTC	56	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
[Languages]	2024-04-25 00:54:49 UTC	2024-04-25 00:54:49 UTC	2024-04-25 00:54:49 UTC	2024-04-25 00:54:46 UTC	416	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
[LICENSE]	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:47 UTC	2024-04-25 00:54:47 UTC	10210	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
[NOTICE]	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	9224	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
Veracrypt.Format.exe	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	6221544	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
veracrypt.cat	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	11903	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
Veracrypt.exe	2024-04-25 00:54:48 UTC	2024-04-25 00:54:53 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	6281064	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
veracrypt.ini	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2880	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
veracrypt.sys	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	813112	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
VeracryptExpander.exe	2024-04-25 00:54:48 UTC	2024-04-25 00:54:53 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	5677288	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
License.txt	2024-04-25 00:54:47 UTC	2024-04-25 00:54:47 UTC	2024-04-25 00:54:47 UTC	2024-04-25 00:54:47 UTC	38143	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt
Veracrypt.Setup.exe	2024-04-18 18:37:43 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	2024-04-25 00:54:48 UTC	35282182	Allocated	Allocated	unknown	/img_winimage.001/vol_vcd3/Program Files/Veracrypt

The bottom pane shows the file's metadata, including its name, size, and location. The file is identified as 'Veracrypt.exe' and is located in the 'Program Files' folder of the 'winimage.001' image.

Figure 7: VeraCrypt Installer found in Documents Folder

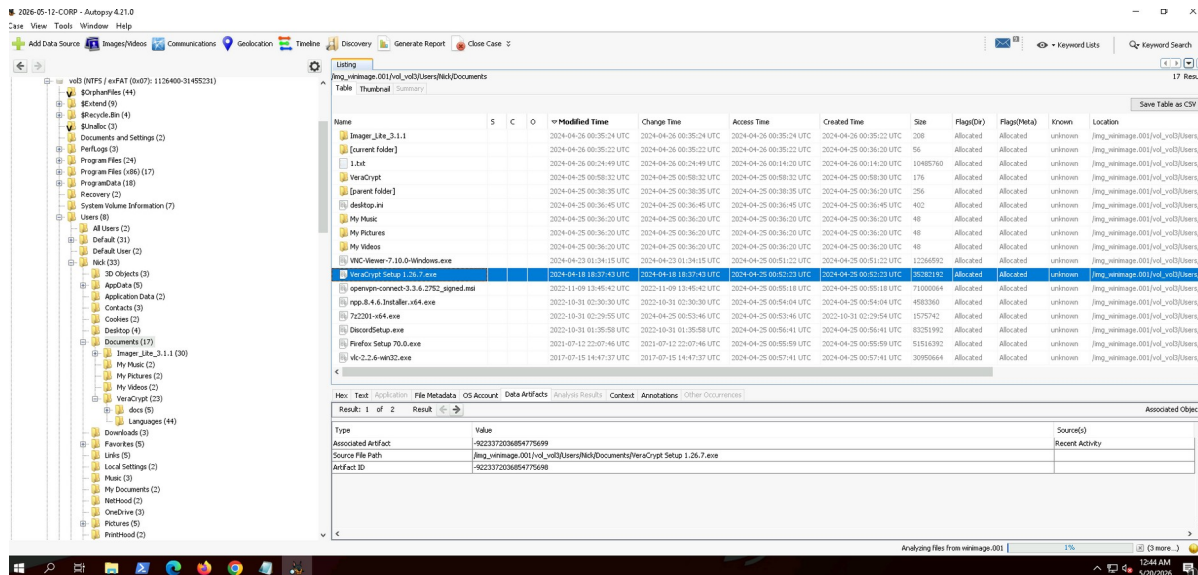


Figure 8: SYSTEM Registry Hive confirming VeraCrypt volumes mounted as drives K:.

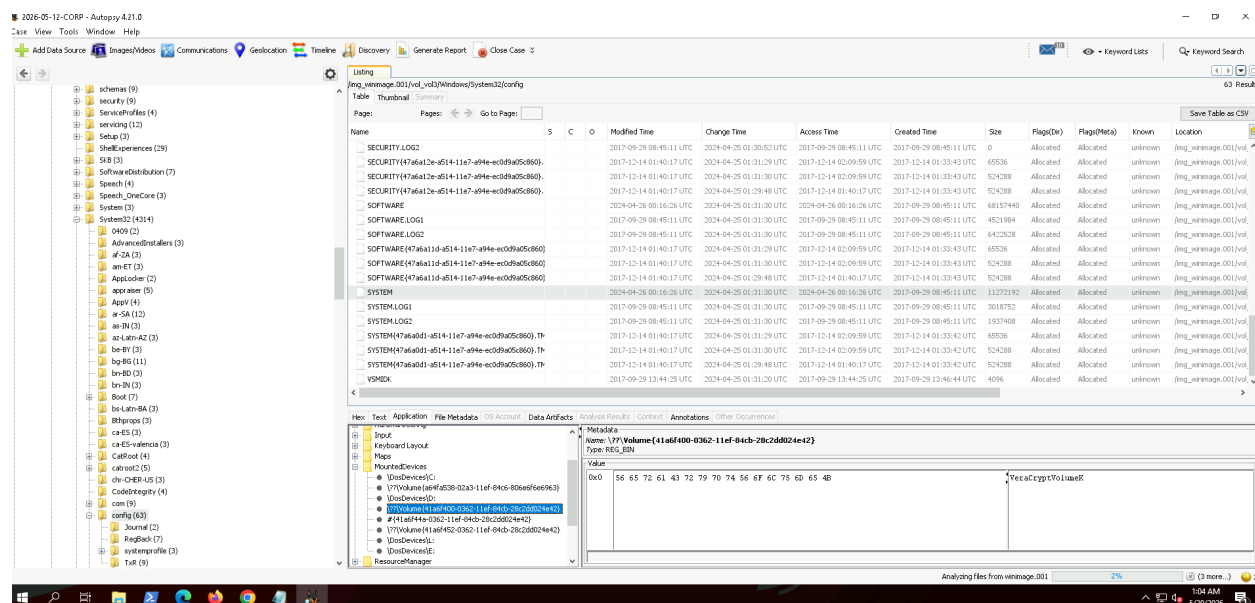


Figure 9: SYSTEM Registry Hive confirming VeraCrypt volumes mounted as drives L:.

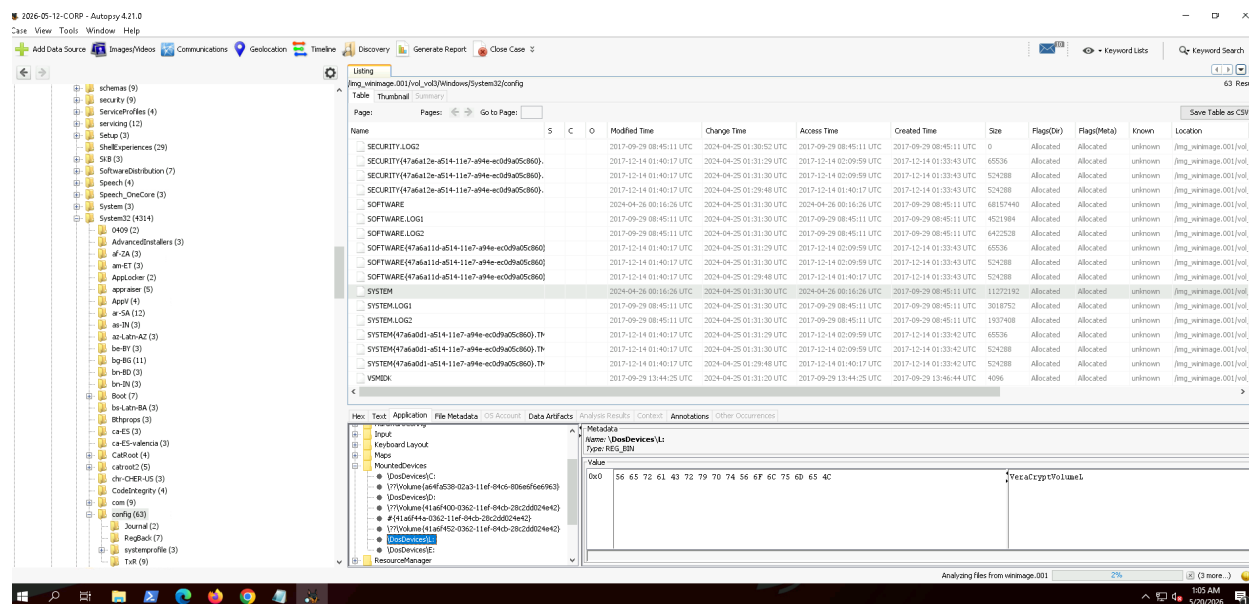
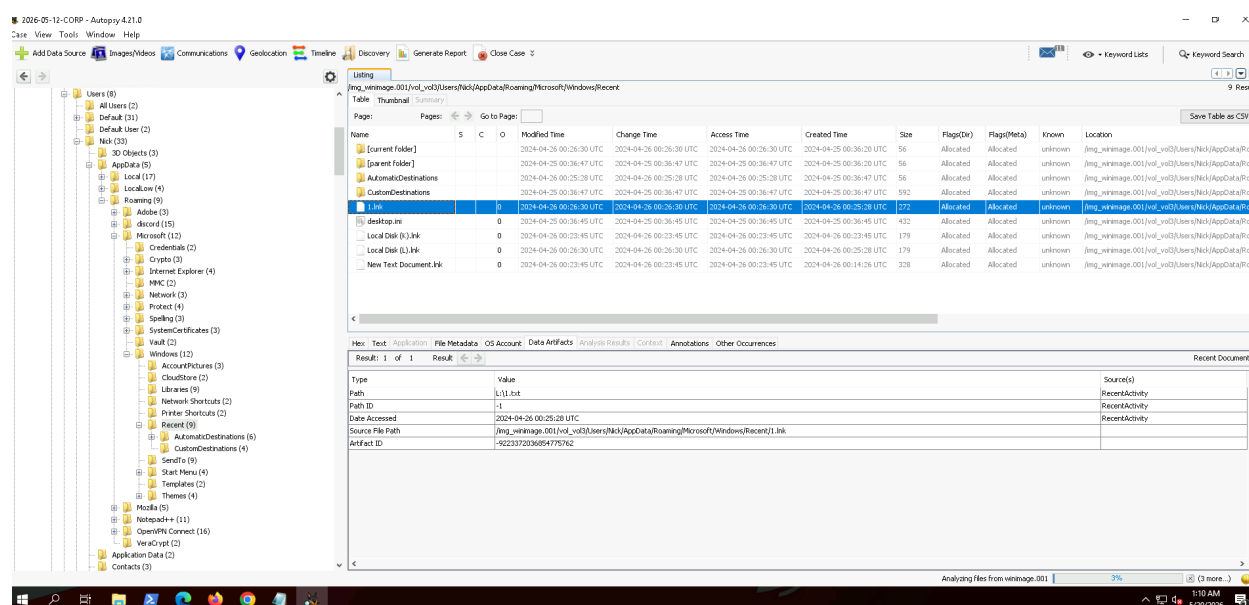
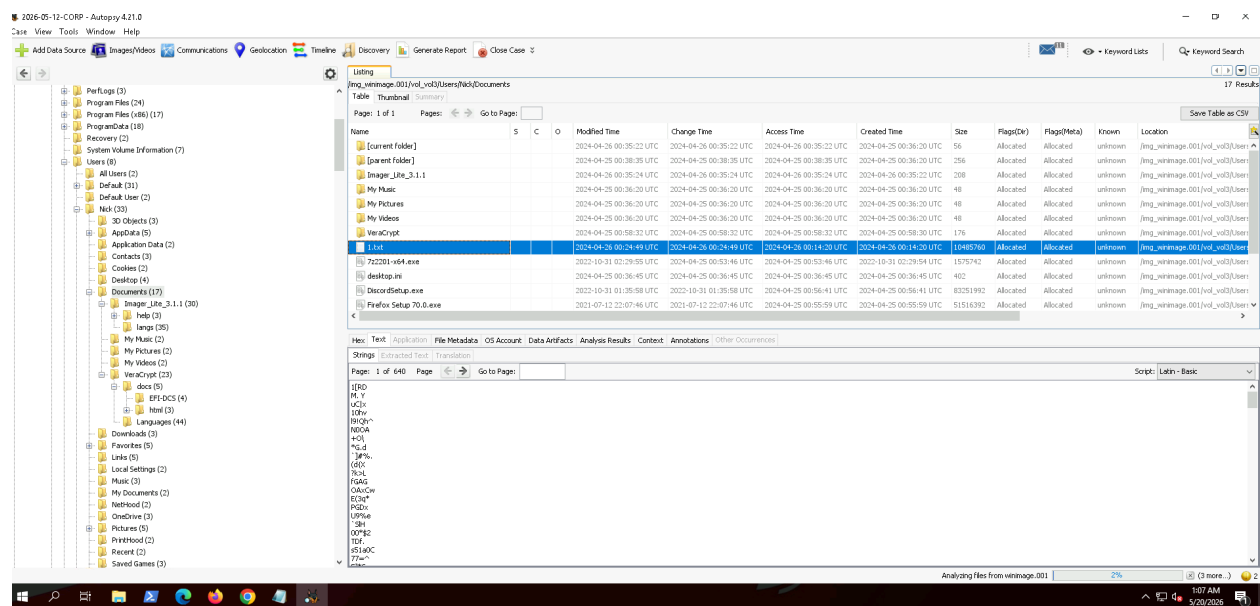


Figure 10: Windows LNK files showing navigation of mounted encrypted drives L and K, as well as possible “1.lnk” for “1.txt”.





Step 5: Automated Artifact Analysis (Hard Drive)

Upon completion of the Autopsy ingest modules on the system hard drive (winimage.001), automated artifact analysis successfully corroborated the manual findings.

Windows Prefetch analysis (C:\Windows\Prefetch) confirmed that VERACRYPT.EXE was executed three times by the user. Furthermore, Web History artifacts revealed that the user 'Nick' utilized a browser application to actively open the 1.txt file located within the mounted encrypted L: drive (file:///L:/1.txt).

Figure 12: Windows Prefetch artifacts confirming the active execution of VeraCrypt.exe and FTK Imager.exe.

The screenshot shows the Autopsy 4.21.0 interface. The main window displays a list of prefetch files. The file VERACRYPT.EXE-047EFD06.pf is highlighted. The bottom pane shows the details for this file, including its type (VERACRYPT.EXE), path (PROGRAM FILES\VERACRYPT), and source (Windows Prefetch Analyzer).

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size	Flags(Dr)	Flags(Meta)	Known	Location
VERACRYPT.EXE-047EFD06.pf				2024-04-26 00:36:51 UTC	2024-04-26 00:36:51 UTC	2024-04-26 00:14:38 UTC	2024-04-26 00:14:38 UTC	11870	Allocated	Allocated	unknown	img_winimage.001\vol_v
Current folder												
MMC.EXE-CF935122.pf				2024-04-26 00:35:55 UTC	2024-04-26 00:35:55 UTC	2024-04-26 00:35:55 UTC	2024-04-26 00:35:55 UTC	15748	Allocated	Allocated	unknown	img_winimage.001\vol_v
VDS.EXE-6E7946F9.pf				2024-04-26 00:35:55 UTC	2024-04-26 00:35:55 UTC	2024-04-26 00:35:55 UTC	2024-04-26 00:35:55 UTC	7223	Allocated	Allocated	unknown	img_winimage.001\vol_v
VOLDR.EXE-8089080B.pf				2024-04-26 00:35:54 UTC	2024-04-26 00:35:54 UTC	2024-04-26 00:35:54 UTC	2024-04-26 00:35:54 UTC	4077	Allocated	Allocated	unknown	img_winimage.001\vol_v
FTV IMAGER.EXE-ED6A583.pf				2024-04-26 00:35:41 UTC	2024-04-26 00:35:41 UTC	2024-04-26 00:35:41 UTC	2024-04-26 00:35:41 UTC	23281	Allocated	Allocated	unknown	img_winimage.001\vol_v
SMARTSCREEN.EXE-985E4173.pf				2024-04-26 00:35:39 UTC	2024-04-26 00:35:39 UTC	2024-04-25 00:34:53 UTC	2024-04-25 00:34:53 UTC	10250	Allocated	Allocated	unknown	img_winimage.001\vol_v
DLHOST.EXE-766398D2.pf				2024-04-26 00:35:36 UTC	2024-04-26 00:35:36 UTC	2024-04-25 00:36:25 UTC	2024-04-25 00:36:25 UTC	7566	Allocated	Allocated	unknown	img_winimage.001\vol_v
SEARCHFILTERHOST.EXE-77482312.pf				2024-04-26 00:35:32 UTC	2024-04-26 00:35:32 UTC	2024-04-25 00:39:00 UTC	2024-04-25 00:39:00 UTC	3994	Allocated	Allocated	unknown	img_winimage.001\vol_v
SEARCHPROTOCOLHOST.EXE-0C8C4DE.pf				2024-04-26 00:35:32 UTC	2024-04-26 00:35:32 UTC	2024-04-25 00:39:00 UTC	2024-04-25 00:39:00 UTC	4519	Allocated	Allocated	unknown	img_winimage.001\vol_v
CONSENT.EXE-5318D6EA.pf				2024-04-26 00:35:31 UTC	2024-04-26 00:35:31 UTC	2024-04-25 00:39:31 UTC	2024-04-25 00:39:31 UTC	42212	Allocated	Allocated	unknown	img_winimage.001\vol_v
RUNDLL32.EXE-CF0EC82C.pf				2024-04-26 00:35:19 UTC	2024-04-26 00:35:19 UTC	2024-04-25 00:51:10 UTC	2024-04-25 00:51:10 UTC	4105	Allocated	Allocated	unknown	img_winimage.001\vol_v
AUDIODG.EXE-B0FD3029.pf				2024-04-26 00:35:17 UTC	2024-04-26 00:35:17 UTC	2024-04-25 00:34:53 UTC	2024-04-25 00:34:53 UTC	6679	Allocated	Allocated	unknown	img_winimage.001\vol_v
CONHOST.EXE-1F8E907E.pf				2024-04-26 00:34:19 UTC	2024-04-26 00:34:19 UTC	2024-04-25 00:40:07 UTC	2024-04-25 00:40:07 UTC	5671	Allocated	Allocated	unknown	img_winimage.001\vol_v
WPCHEMUIB.EXE-4010B4.pf				2024-04-26 00:34:19 UTC	2024-04-26 00:34:19 UTC	2024-04-26 00:27:02 UTC	2024-04-26 00:27:02 UTC	7681	Allocated	Allocated	unknown	img_winimage.001\vol_v
TIMORIER.EXE-0223467C.pf				2024-04-26 00:32:13 UTC	2024-04-26 00:32:13 UTC	2024-04-25 00:56:35 UTC	2024-04-25 00:56:35 UTC	7188	Allocated	Allocated	unknown	img_winimage.001\vol_v
TRUSTEDINSTALLER.EXE-3CC531E5.pf				2024-04-26 00:32:13 UTC	2024-04-26 00:32:13 UTC	2024-04-25 00:56:35 UTC	2024-04-25 00:56:35 UTC	4416	Allocated	Allocated	unknown	img_winimage.001\vol_v

Details for VERACRYPT.EXE-047EFD06.pf:

Type	Value	Source(s)
Program Name	VERACRYPT.EXE	Windows Prefetch Analyzer
Path	PROGRAM FILES\VERACRYPT	Windows Prefetch Analyzer
CreateTime	2024-04-26 00:36:50 UTC	Windows Prefetch Analyzer
Count	3	Windows Prefetch Analyzer
Comment	Prefetch File	Windows Prefetch Analyzer
Source File Path	img_winimage.001\vol_v\Windows\Prefetch\VERACRYPT.EXE-047EFD06.pf	
Artifact ID	922337026854774455	

Figure 13: Web History artifact showing the suspect opening the encrypted 1.txt file from the mounted L: drive.

The screenshot displays the Autopsy 4.21.0 interface. The left sidebar shows a tree view of artifacts, with 'Web History' selected under 'Data Artifacts'. The main pane shows the 'Web History' artifact details. The 'Table' tab is active, displaying a list of web history entries. The selected entry is highlighted in blue.

Source Name	S	C	O	URL	Date Accessed	Program Name	Domain	Username	Data Source
WebCacheV01.dat				file:///C:/Users/Nick/Documents/Temp/2017ext%20Docume...	2024-04-26 00:14:26 UTC	Microsoft Edge Analyzer		Nick	hennpage.001
WebCacheV01.dat				file:///L:/1.txt	2024-04-26 00:26:30 UTC	Microsoft Edge Analyzer		Nick	hennpage.001

Below the table, the 'Hex' tab is active, showing the extracted text for the selected entry. The text is displayed in a monospace font.

URL : file:///L:/1.txt
Date Accessed : 2024-04-26 00:26:30 UTC
Program Name : Microsoft Edge Analyzer
Domain :
Username : Nick

Step 6: Volatile Memory Processing and Volatility Analysis (Item 003)

The volatile memory capture Windows 10 x64-31acf216.vmem was ingested into the forensic workstation under read-only, software write-blocked conditions to preserve data integrity. The memory was analyzed using Volatility 2.6. The imageinfo plugin successfully identified the operating system parameters. However, attempts to execute structured process and network plugins (pstree, pslist) resulted in fatal Page Fault errors and Python execution exceptions. This structural degradation is a direct result of "memory smear" introduced by local IT personnel during live acquisition and delayed archiving.

Figure 14: Volatility 2.6 Page Fault Screenshot

```

Select Administrator: Command Prompt

C:\Users\Administrator>cd C:\User

C:\Users\Administrator>cd C:\Program Files\CyberTools\volatility

C:\Program Files\CyberTools\volatility>volatility -f "C:\Users\Administrator\Desktop\Lab Resources\Windows 10 x64-31acf216.vmem" --profile=win10x64_18586 pslist
Volatility Foundation Volatility Framework 2.6
Offset(P)      Name          PID  PPID PDB          Time created      Time exited
-----
WARNING : volatility.debug : Cannot find nt!ObGetObjectType
WARNING : volatility.debug : Cannot find nt!ObGetObjectType
Traceback (most recent call last):
  File "vol.py", line 192, in <module>
  File "vol.py", line 183, in main
  File "volatility\commands.py", line 147, in execute
  File "volatility\plugins\filescan.py", line 423, in render_text
  File "volatility\poolscan.py", line 252, in scan
  File "volatility\plugins\overlays\windows\windows.py", line 1141, in get_object
  File "volatility\plugins\overlays\windows\windows.py", line 1114, in get_object_top_down
  File "volatility\plugins\overlays\windows\win7.py", line 155, in get_object_type
  File "volatility\plugins\overlays\windows\win10.py", line 272, in TypeIndex
TypeError: unsupported operand type(s) for ^: 'long' and 'NoneType'
Failed to execute script vol

C:\Program Files\CyberTools\volatility> Cannot find nt!ObGetObjectType_

```


Step 7: Raw Data Carving and Cryptographic Key Extraction (Item 003)

Due to the structural corruption identified in Step 6, the analysis pivoted to raw, unstructured data carving using Bulk Extractor and Autopsy (ingested as an Unallocated Space Image). This approach bypassed the corrupted operating system structures. Bulk Extractor successfully carved several candidate AES-256 master key schedules from the raw memory space. These keys were physically verified in Autopsy's Hex viewer at their respective decimal offsets (such as Offset 180503280). Additionally, keyword carving recovered raw execution strings showing VeraCrypt.exe running concurrently with Windows Disk Management (diskmgmt.msc), proving the suspect was manipulating disk partitions to mount encrypted containers.

Figure 15: *aes_keys.txt* from Carved Directory:

```

aes_keys.txt - Notepad
File Edit Format View Help
# BANNER FILE NOT PROVIDED (-b option)
# BULK_EXTRACTOR-Version: 1.5.2 ($Rev: 10844 $)
# Feature-Recorder: aes_keys
# Filename: C:\Users\Administrator\Desktop\Lab Resources\Windows 10 x64-31acf216.vmem
# Feature-File-Version: 1.1
175760048 5c 10 90 96 ed ff 5d 6f c9 01 5b be bd 24 4f dc AES128
180503280 07 df 0d 6f 64 30 a9 96 fb 0f 64 6f ae 92 2a 31 52 0b 1a d1 4e 5f ef 99 da 54 ba ff 8c ab 16 c3 AES256
180503744 07 df 0d 6f 64 30 a9 96 fb 0f 64 6f ae 92 2a 31 52 0b 1a d1 4e 5f ef 99 da 54 ba ff 8c ab 16 c3 AES256
180504400 00 01 02 03 04 05 06 07 08 09 0a 0b 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d 1e 1f AES256
505795757 4c af 81 5e f9 94 05 8f b6 43 b2 c0 7d 0f 89 22 AES128
791667357 a5 95 a7 72 97 db 14 02 14 e6 ea 33 24 e0 16 8d 22 b1 5e f9 f8 08 50 78 87 cb b6 d7 eb 13 ee c6 AES256
905574365 6e 98 5f ac f6 40 de 89 bd ae 80 63 5d 68 df 5f AES128
1774051373 da 8d 6a 87 03 a0 71 79 ac 2c d4 fe df bd 2c 4a 51 67 0d ca 3f 59 c9 9d 9b 50 a4 61 df a2 27 7b AES256
1774052029 c7 b7 0e 7a b9 19 e9 b3 20 14 87 d4 fd 42 cb 8b AES128
1774052685 ac ea 40 c3 98 12 db 98 e2 00 bd ff ef 6d 6b 4b AES128
1779298685 6e 21 5f 93 5e 95 f7 21 79 d0 cb 70 ba 59 72 a4 35 02 03 67 af cb b3 6c 3f 1f 8d e4 6e 8a 92 91 AES256
1779299341 68 22 d0 bb f4 04 1d 6a 84 64 c6 6e 42 b4 b3 01 07 f0 c0 61 ae a6 a4 3b 13 92 84 45 a1 84 64 a1 AES256
2114382845 27 67 cb 3c 64 79 e8 70 9b 3c d6 d8 07 70 6b 83 AES128

```

Figure 16: Autopsy Hex view showing the carved AES-256 master key at offset 180503280:

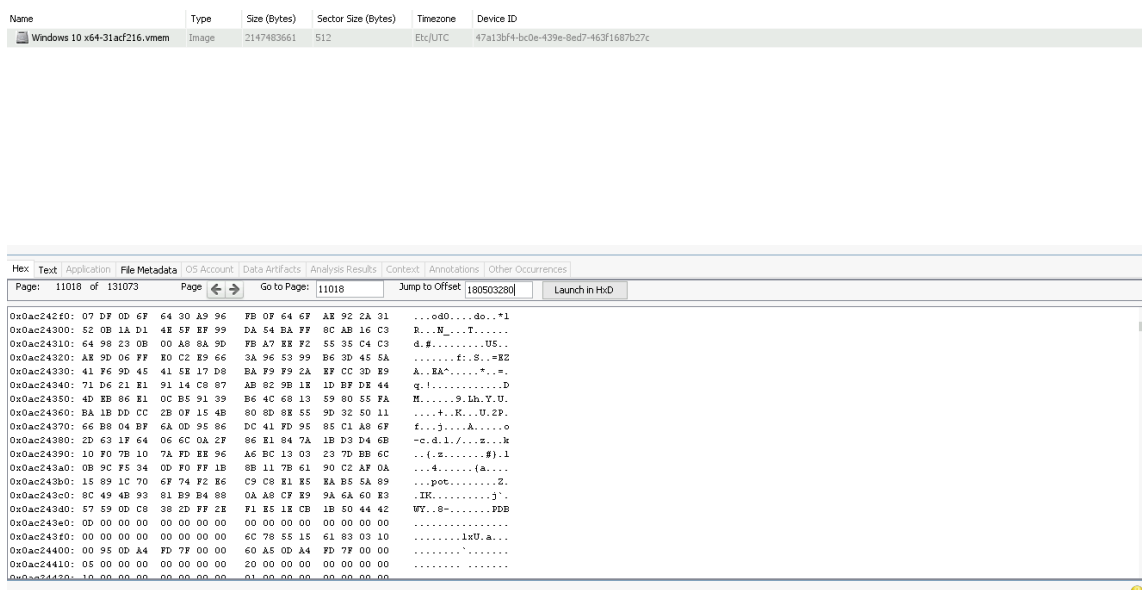
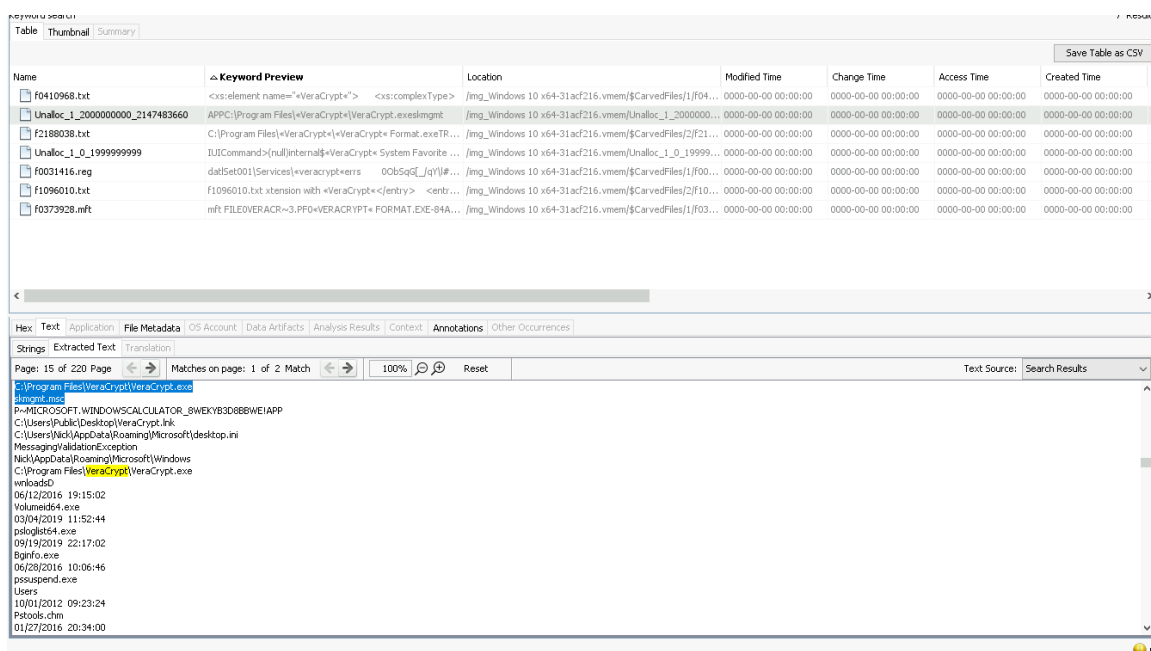


Figure 17: Autopsy raw string carve showing VeraCrypt running concurrently with diskmgmt.msc.



Findings

The combined evidence from the USB drive (Item 001), system hard drive (Item 002), and the system memory capture (Item 003), provides a complete behavioral/technical timeline of the insider threat. Nicholas Anderson, disgruntled over a denied transfer and poor performance reviews, utilized corporate systems to exfiltrate database records outside of standard operating hours.

To hide this exfiltration, Anderson installed and executed VeraCrypt. The hard drive registry keys and Recent LNK shortcuts prove that Anderson mounted encrypted containers as drives `K:` and `L:`. The volatile memory carving corroborated this activity by recovering active RAM execution strings for `VeraCrypt Format.exe` and direct file paths to the mounted `L:/1.txt` volume. The recovery of these AES-256 keys provides the raw cryptographic material that are necessary for attempting direct volume decryption in the next phase of this investigation.

Missing Results and Pending Questions

All previous questions regarding the missing memory capture and encrypted containers have been resolved. The IT department recovered the memory dump; however, their improper acquisition and archiving techniques introduced severe memory smearing, rendering standard Volatility process-mapping unfeasible.

A pivot to raw data carving bypassed this issue. The AES-256 master encryption keys were successfully extracted from raw memory. The pending phase of this investigation will involve

using the master keys to mount the carved container files (`stuff.txt` and `1.txt`) to extract, review, and catalog the exfiltrated database files.

Table 3: File Names, Types, Source, and Relevance

File Name	Type	Source	Relevance
Email 2 Oct 1.pdf	Document	Evidence 001 (flash drive)	Sign of off-hours database access and anger over a lost transfer.
Memo Oct 5.pdf	Document	Evidence 001	Company databases contain highly sensitive client credit/debit card information.
Perf Oct 20.pdf	Document	Evidence 001	HR Performance Improvement Plan warning.
Resignation Oct 21.pdf	Document	Evidence 001	Confirmed identity: Nicholas Anderson. Shows hostile attitude.
stuff.txt	.txt	Evidence 001	Unstructured ciphertext with an entropy of 7.999982.
1.txt	.txt	Evidence 002 (Windows Disk Image)	Suspected secondary 10MB encrypted container located in Documents folder.
VeraCrypt Setup & Folders	Executable/Dir	Both 001 & 002	Identifies the encryption software. Found on both USB and Hard Drive.

File Name	Type	Source	Relevance
SYSTEM (MountedDevices)	Registry Hive	002	Proves VeraCrypt volumes were actively mounted as drives K: and L:
Local Disk (K).lnk	LNK File	002	Proves the suspect actively navigated inside the mounted encrypted volumes.
netfx1.cab & .bmps	Archives	001	Potential secondary hiding methods (steganography) requiring review.
7z2201-x64.exe	Executable	002	7-Zip installer; potential staging tool used to compress stolen data.
Imager_Lite_3.1.1	Directory	002	IT forensic tool. Likely source of the missing memory capture.
VERACRYPT.EXE[...].pf	Prefetch file	002	Proves Veracrypt was executed 3 times on the corporate machine.
FTK Imager.exe-[...].pf	Prefetch file	002	Proves IT personnel execute the RAM capture.
Web History (file:///L:/1.txt)	Web Artifact	002	Proves that suspect actively opened the 1.txt file inside the mounted L: drive.

File Name	Type	Source	Relevance
'aes_keys.txt' (Offsets 180503280, etc.)	RAM Fragment	003	Carved AES-256 master keys matching VeraCrypt's active RAM footprint.
VeraCrypt_XML('f 0410968.txt')	RAM Fragment	003	Carved VeraCrypt localization file, proving the GUI was active in memory.

References

National Institute of Standards and Technology. (2022, September 28). Digital evidence preservation: Considerations for evidence handlers. NIST.
<https://nvlpubs.nist.gov/nistpubs/ir/2022/NIST.IR.8387.pdf>